TAMIL NADU STATE BOARD - STANDARD 9 SCIENCE (PHYSICS)

CHAPTER 3: FORCE AND LAWS OF MOTION

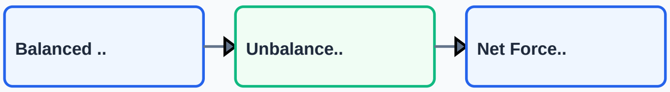
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 9 English Medium
Subject:	Science (Physics)
Chapter Name:	Force and Laws of Motion

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

TOPIC ROADMAP: FORCE AND LAWS OF MOTION



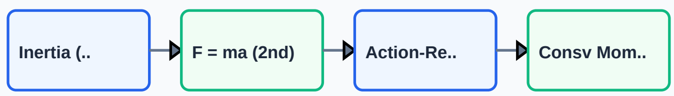
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Inertia	The inherent property of a body to resist any change in its state of rest or uniform motion in a straight line, unless acted upon by an external force.
Momentum	The quantity of motion contained in a body, measured as the product of its mass and velocity. SI unit: kg m/s.
Impulse	A large force acting on a body for a very short interval of time, equal to the change in momentum. SI unit: N s or kg m/s.
Balanced Forces	When a set of forces acting on a body does not change its state of rest or uniform motion, yielding a net force of zero.
Friction	The opposing force that comes into play at the contact surface when one body slides or rolls over another.

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

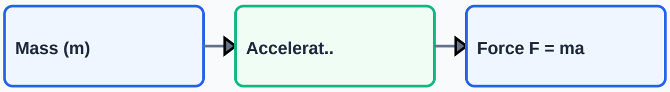
GLOSSARY: NEWTON'S LAWS FORCE VECTORS BALANCE



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

FORMULA: NEWTON'S SECOND LAW CALCULATION



Essential Formulas, Laws & Identities:

- Linear Momentum: $p = m * v$
- Newton's Second Law: Force (F) = mass (m) * acceleration (a)
- Impulse (J) = Force (F) * time (t) = change in momentum ($m*v - m*u$)
- Law of Conservation of Momentum: $m1*u1 + m2*u2 = m1*v1 + m2*v2$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: A force of 15 N acts on a body of mass 3 kg. Find the acceleration produced.

Step-by-step Solution:

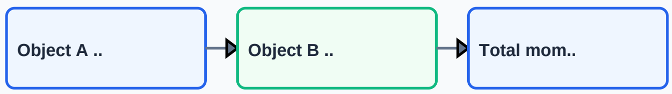
Given: F = 15 N, m = 3 kg. Using F = m * a: 15 = 3 * a => a = 5 m/s².

Example 2: Calculate the change in momentum of a 0.5 kg ball moving at 20 m/s that is hit back along the same path at 15 m/s.

Step-by-step Solution:

Given: m = 0.5 kg, u = 20 m/s, v = -15 m/s (opposite direction). Change in momentum = m*v - m*u = 0.5 * (-15) - 0.5 * 20 = -7.5 - 10 = -17.5 kg m/s.

WORKED: CONSERVATION OF MOMENTUM IN COLLISION



HOTS (Higher Order Thinking Skills) Challenge

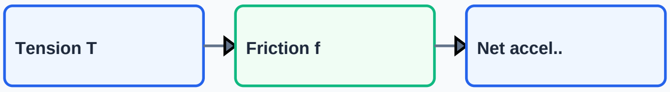
Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Why does a high jumper land on a cushion or a sand bed rather than concrete floor?

Analytical Solution Strategy:

By landing on a cushion or sand, the time interval of impact (t) increases significantly. Since Impulse (change in momentum) is constant, increasing the duration decreases the average force (F = J / t) exerted on the athlete's body, preventing severe fractures.

HOTS: BALANCED VECTORS ON INCLINED PLANE



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): State Newton's three laws of motion with daily life examples.

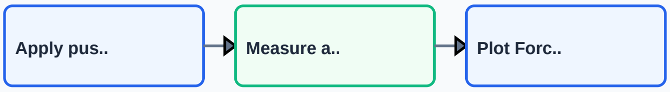
Q2 (2-Mark): Define 1 Newton of force.

Q3 (2-Mark): Show that Newton's First Law is also called the Law of Inertia, explaining its three types.

Q4 (5-Mark): A bullet of mass 20 g is fired with a velocity of 150 m/s from a pistol of mass 2 kg. Calculate the recoil velocity of the pistol.

Q5 (5-Mark): State and prove the Law of Conservation of Momentum.

PRACTICE: FRICTION COEFFICIENT WORKSHEET



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Safety Equipment

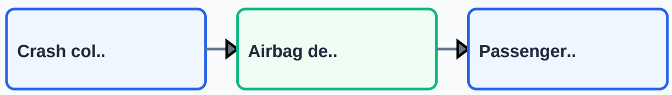
Vehicular crumple zones, seat belts, and airbags are designed to collapse slowly during collisions, lowering impact forces by extending time.

Application Case: Athletic Shoe Design

Sneakers incorporate gel or foam soles to reduce peak impact forces on joints during running.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

APPLICATION: VEHICLE SAFETY CRASH TESTS ANALYSIS



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.

✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.

✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3, N) for the final numeric answer.

✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

[ ] Read and memorized core definitions on Page 2.

[ ] Reviewed critical formulas and relations on Page 3.

[ ] Solved worked examples on Page 4.

[ ] Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress. Good luck!

REVISION: FORCE DYNAMICS FLOWCHART